

Q2. 证: 叠加原理

设 $x_1(t), x_2(t)$ 为非齐次线性方程的解

$$\begin{cases} \frac{d^{n-1}x}{dt^{n-1}} + a_1(t) \frac{d^{n-2}x}{dt^{n-2}} + \dots + a_n(t)x = f_1(t) \\ \frac{d^{n-1}x}{dt^{n-1}} + a_1(t) \frac{d^{n-2}x}{dt^{n-2}} + \dots + a_n(t)x = f_2(t) \end{cases}$$

则: $x_1(t) + x_2(t)$ 是方程.

$$\left[\frac{d^{n-1}(x_1+x_2)}{dt^{n-1}} + a_1(t) \frac{d^{n-2}(x_1+x_2)}{dt^{n-2}} + \dots + a_n(t)(x_1+x_2) \right] = f_1(t) + f_2(t)$$

①
算子

$$L(x) = \frac{d^{n-1}x}{dt^{n-1}} + a_1(t) \frac{d^{n-2}x}{dt^{n-2}} + \dots + a_n(t)x$$

(n 阶线性微分算子)

②
线性

$$\begin{cases} L[cu] = cL[u], \quad c \in \mathbb{R} \\ L[u_1 + u_2] = L[u_1] + L[u_2] \end{cases}$$

③
已知条件

$$\begin{cases} L[x_1] = f_1(t) \\ L[x_2] = f_2(t) \end{cases}$$

④ 验证和函数性质

$$\begin{aligned} L[x_1(t) + x_2(t)] &= L[x_1(t)] + L[x_2(t)] \\ &= f_1(t) + f_2(t) \end{aligned}$$

⑤

当 $x_1(t) + x_2(t)$ 作用于线性微分算子 L , 其输出结果恰如等于两原函数是非齐次项代数和 $f_1(t) + f_2(t)$

$$\text{证毕: } L[x] = f_1(t) + f_2(t)$$

$$x = x_1 + x_2$$

~~证毕~~ 满足叠加原理

Q4 $\frac{d^2x}{dt^2} - x = 0$, 基函数组 e^t, e^{-t}

$$\begin{cases} \textcircled{1} x(0) = 1, x'(0) = 0 \\ \textcircled{2} x(0) = 0, x'(0) = 1 \end{cases}$$

求初值条件如上, 的基函数组, ($W(0)=1$)

① ~~通解~~: $x(t) = c_1 e^t + c_2 e^{-t} \xrightarrow{t=0, x(0)=\square} x(0)=\square \rightarrow \varphi_1$
 $x(t) = c_1 e^t - c_2 e^{-t} \xrightarrow{t=0, x'(0)=\square} x'(0)=\square \rightarrow \varphi_2$
 两基函数组

② 标准基函数组 (原书)

(A) $\begin{cases} x(0) = 1 \\ x'(0) = 0 \end{cases} \Rightarrow \begin{cases} 1 = c_1 + c_2 \\ 0 = c_1 - c_2 \end{cases}$

2组解 $\Rightarrow$ 积分
(+ 每次应用)

$c_1 = c_2, c_1 = c_2 = \frac{1}{2}$

$\therefore$ 标准基函数 $\varphi_1(t) = \frac{e^t + e^{-t}}{2} = \cosh t$

$\begin{cases} x(t) \\ x'(t) \end{cases}$
再解 (A) φ_1, φ_2

(B) $\begin{cases} x(0) = 0 \\ x'(0) = 1 \end{cases} \Rightarrow \begin{cases} 0 = c_1 + c_2 \\ 1 = c_1 - c_2 \end{cases} \Rightarrow \begin{cases} c_1 = -c_2 \\ 1 = -2c_2 \end{cases} \Rightarrow \begin{cases} c_1 = \frac{1}{2} \\ c_2 = -\frac{1}{2} \end{cases}$

$\varphi_2(t) = \frac{e^t - e^{-t}}{2} = \sinh t$

标准基函数组 $W(0)=1$

φ_1, φ_2 构成基, $t=0, W(0) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ (标准基函数)

$W(t) = \begin{vmatrix} \varphi_1 & \varphi_2 \\ \varphi_1' & \varphi_2' \end{vmatrix} = \varphi_1 \varphi_2' - \varphi_1' \varphi_2 = 1$

③ 求初值条件 $x(0) = x_0, x'(0) = x_0'$

$x(t) = x_0 \varphi_1(t) + x_0' \varphi_2(t)$

$x(t) = x_0 \frac{e^t + e^{-t}}{2} + x_0' \frac{e^t - e^{-t}}{2}$
 $= x_0 \cosh t + x_0' \sinh t$

标准解

Q6 $x_1(t) \neq 0$ 二阶齐次线性微分方程

★ $\begin{cases} x'' + a_1(t)x' + a_2(t)x = 0, \text{ 对 } x \\ a_1(t), a_2(t) \text{ 于 } [a, b] \text{ 连续} \end{cases}$

证: (1) $x_1(t)$ 为方程的充要条件

$$W'[x_1, x_2] + a_1 W[x_1, x_2] = 0,$$

$$\Rightarrow W' + a_1 W = 0,$$

$$\because W = x_1 x_2' - x_1' x_2, \quad W' = (x_1 x_2' - x_1' x_2)' = x_1 x_2'' - x_1'' x_2$$

~~$$x_1 x_2'' = -a_1 x_1' x_2 - a_2 x_1 x_2$$~~

若 x_1, x_2 均为方程的解 $(x'' + a_1 x' + a_2 x = 0)$

$$\begin{cases} x_1'' + a_1 x_1' + a_2 x_1 = 0 \\ x_2'' + a_1 x_2' + a_2 x_2 = 0 \end{cases}$$

$$\begin{aligned} \text{代入 } W' &= -x_1(a_1 x_2' + a_2 x_2) + x_2(a_1 x_1' + a_2 x_1) \\ &= -a_1 W \end{aligned}$$

$$W' + a_1 W = 0 \quad (\text{D&F})$$

(2) 通解表达式

$$W' + a_1(t)W = 0,$$

$$W(t) = C e^{(-\int_{t_0}^t a_1(s) ds)}$$

$$\frac{d}{dt}\left(\frac{x_2}{x_1}\right) = \frac{x_1 x_2' - x_1' x_2}{x_1^2} = \frac{W(t)}{x_1^2}$$

$$\frac{x_2}{x_1} = \int \frac{1}{x_1^2} e^{-(\int_{t_0}^t a_1(s) ds)} dt + C_2$$

$$x = C_1 x_2 + C_2 x_1, \quad (\text{通解} = \text{特解} + \text{齐次解})$$

$$x = x_1 \left[C_1 \int \frac{1}{x_1^2} e^{(-\int_{t_0}^t a_1(s) ds)} dt + C_2 \right]$$

例 4.2

Q2 (1) $x^{(4)} - 5x'' + 4x = 0$,

特征: $\lambda^4 - 5\lambda^2 + 4 = 0$

$(\lambda^2 - 1)(\lambda^2 - 4) = 0$

$\begin{cases} \lambda_{1,2} = \pm 1 \\ \lambda_{3,4} = \pm 2 \end{cases}$

通解: $x(t) = c_1 e^t + c_2 e^{-t} + c_3 e^{2t} + c_4 e^{-2t}$, $c_i \in \mathbb{R}$

(2) $x''' - 3ax'' + 3a^2x' - a^3x = 0$

特征: $\lambda^3 - 3a\lambda^2 + 3a^2\lambda - a^3 = 0$

$(\lambda - a)^3 = 0 \Rightarrow \lambda = a$ (三重根)

通解: $x(t) = e^{at}(c_1 + c_2 t + c_3 t^2)$ 常系数线性齐次方程

齐次方程以特征根为基

(3) $x^{(5)} - 4x^{(3)} = 0$

特征: $\lambda^5 - 4\lambda^3 = 0$

$\lambda^3(\lambda^2 - 4) = 0$,

$\begin{cases} \lambda = 0 & (\text{三重根}) \\ \lambda = \pm 2 \end{cases}$

通解: $x(t) = \underbrace{(c_1 + c_2 t + c_3 t^2)}_{\text{齐次解}} + (c_4 e^{2t} + c_5 e^{-2t})$

(4) $x'' + x' + x = 0$

特征: $\lambda^2 + \lambda + 1 = 0$,

$\lambda = \frac{-1 \pm \sqrt{3}i}{2}$ (复根)

$\alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2}$

通解: $x(t) = e^{-\frac{1}{2}t} (c_1 \cos \frac{\sqrt{3}}{2}t + c_2 \sin \frac{\sqrt{3}}{2}t)$